Malak Serag Ibrahim

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Summary

4th-year Computer Engineering student with hands-on experience across software development, DevOps, cybersecurity, cloud computing, data analytics, and business intelligence. Built full-stack applications, CI/CD pipelines, containerized services, security-focused systems, and data-driven applications through internships and academic projects. Experienced with Linux, Git/GitLab, Docker, Python, Java, JavaScript, and modern development tools, with a strong foundation in systems, networking, and computer engineering.

Experience

DevOps Intern — Aug 2026 – Sep 2026
MISR TECHNOLOGY SERVICES (MTS) — *Cairo, Egypt*

- Worked hands-on with Linux/Ubuntu, Git/GitLab, GitLab CI/CD, GitLab Runner, Maven, Docker, Docker Hub, PostgreSQL, Spring Boot, and VirtualBox in a practical DevOps environment.
- Designed and implemented a complete CI/CD pipeline for a Spring Boot Student Manager application with automated Build, Test, Package, Docker Image Build, Docker Hub Push, and Deployment stages.
- Configured GitLab Runner using the Shell executor on Ubuntu and integrated CI/CD variables for Docker Hub authentication and automated image delivery.
- Containerized the Spring Boot application and PostgreSQL database using Docker, configuring Docker networks, persistent volumes, port mapping, and service connectivity.
- Deployed the application to an Ubuntu virtual machine through VirtualBox and troubleshot PostgreSQL connectivity, Docker networking, ports, Maven builds, CI/CD execution, and application deployment.

Data Science & Business Intelligence Intern — Aug 2026
MISR TECHNOLOGY SERVICES (MTS) — *Cairo, Egypt*

- Performed exploratory data analysis on large trade datasets using Python, Pandas, Matplotlib, and Seaborn to identify trends, correlations, outliers, and business insights.
- Applied descriptive and inferential statistical analysis, data cleaning, visualization, and correlation analysis to transform raw datasets into meaningful business insights.
- Built interactive Tableau dashboards using KPI cards, calculated fields, filters, date analysis, data modeling, and fact/dimension relationships.
- Worked with BI concepts including Data Warehouses, Data Marts, OLAP, ETL, SSBI, and business KPI development.

Cybersecurity & Cloud Computing Intern — Jul 2026
MIDEAST COMMUNICATION SYSTEMS (MCS) — *Cairo, Egypt*

- Completed a four-week program combining intensive cybersecurity training, practical labs, a technical CTF, and security-team shadowing across enterprise environments.
- Worked with IAM, PAM, Active Directory, BeyondTrust, NDR, IPS, TAP/SPAN, Trellix, vulnerability management, OWASP, email security, and endpoint security concepts.
- Gained practical exposure to Linux/Ubuntu, WSL, Docker, NGINX, virtualization, VMware ESXi/vSphere, cloud computing, enterprise architecture, and disaster recovery concepts.
- Worked with security platforms including Palo Alto Cortex XDR/XSIAM, Kaspersky Cloud, DLP, endpoint security policies, monitoring, and security analytics.
- Passed a three-round cybersecurity CTF covering Network Team, Blue Team, and Red Team scenarios with a 92% overall score.
- Performed practical labs and technical activities involving Docker, Linux administration, security monitoring, vulnerability assessment, Log4Shell/OWASP research, and enterprise security tools.

Full-Stack Developer Intern — Jul 2025 – Aug 2025
MISR TECHNOLOGY SERVICES (MTS) — *Cairo, Egypt*

- Developed full-stack web and mobile applications using Vue.js, Nuxt.js, Node.js, Express.js, and Flutter.
- Built REST APIs, responsive user interfaces, and JSON-based application functionality for news, forum, and blog platforms.
- Contributed to backend API development and frontend implementation, including the Discover tab of a news platform.
- Used Git, Postman, VS Code, Vue DevTools, Tailwind CSS, Android Studio, Flutter DevTools, and related development tools.

IT Trainee — Aug 2025
PETROJET — *Cairo, Egypt*

- Gained practical exposure to enterprise networking, system administration, infrastructure, GNS3, and SAP environments.

Computer Science OL IGCSE Teaching Assistant — Jul 2021 – May 2024
ROWAD COLLEGE — *Cairo, Egypt*

- Supported IGCSE students in programming, algorithms, logic gates, and exam preparation.

- Simplified complex technical concepts and adapted explanations to different learning levels.

Math OL/AS IGCSE Teaching Assistant — Nov 2023 – Feb 2024
ROWAD COLLEGE — *Cairo, Egypt*

- Supported IGCSE students in mathematical problem-solving, worked examples, and exam preparation.

Projects

Student Manager — CI/CD & Docker Deployment — 2026
Java 17 · Spring Boot · Maven · GitLab CI/CD · Docker · Docker Hub · PostgreSQL · Ubuntu

- Developed a Spring Boot Student Manager application with functionality for searching, adding, editing, and deleting student records.
- Designed a GitLab CI/CD pipeline implementing automated Build, Test, Package, Docker Image Build, Push, and Deployment stages.
- Built a multi-stage Docker image for the Spring Boot application and published it to Docker Hub through the CI/CD pipeline.
- Deployed the application and PostgreSQL database as separate Docker containers connected through a dedicated Docker network with persistent database storage.
- Configured GitLab Runner on Ubuntu and implemented automated deployment while troubleshooting application connectivity, PostgreSQL, Docker networking, port mapping, and VirtualBox networking.

GitLab — Docker Hub

Machine Learning & Data Analytics Projects — Mar 2026 – Jun 2026
Python · Pandas · Scikit-learn · Matplotlib · Seaborn

- Implemented Linear Regression, Logistic Regression, and K-Means clustering projects involving data cleaning, scaling, outlier handling, feature engineering, model evaluation, and visualization.
- Performed exploratory data analysis and statistical analysis on multiple datasets to identify patterns, relationships, trends, and meaningful analytical insights.
- Worked with real-world datasets including housing data, World Happiness data, NBA statistics, and MOOC/course-related data using Python-based analytics and visualization techniques.

Housing — Happiness — NBA — MOOC NLP

AES-128 Encryption Simulator — Jun 2026
Python · Tkinter · Cryptography

- Implemented AES-128 encryption from scratch, including SubBytes, ShiftRows, MixColumns, AddRoundKey, S-Box operations, and key expansion.
- Developed a Tkinter GUI to visualize the AES 4×4 state matrix, encryption stages, and key expansion process.

GitHub: AES Encryption

RFID Scanning & Access Control System — Jun 2026
8051 Assembly · Arduino · MFRC522 · EEPROM · Proteus

- Developed an RFID-based access control system integrating Arduino, MFRC522, AT89C51, EEPROM, LCD, servo motor, LEDs, buzzer, and serial communication.
- Implemented UID verification and access-control logic with communication between the RFID reader, microcontroller, and memory components.
- Designed and tested the complete hardware system through Proteus simulation.

GitHub: RFID System

Cinema Booking System — Dec 2025
Java · JavaFX · Database · Full-Stack Development

- Developed a cinema booking system with a JavaFX graphical interface and database integration for managing movies, showtimes, seats, and reservations.
- Implemented interactive seat selection and booking functionality with database-driven reservation management.
- Designed application logic for adding, updating, deleting, and retrieving cinema and booking data while maintaining consistency between the interface and database.

GitHub: Cinema Booking

News Platform — Web & Mobile — Jul 2025
Vue.js · Nuxt.js · Node.js · Express.js · Flutter · Tailwind CSS

- Developed a full-stack news platform with web and mobile applications using Vue.js, Nuxt.js, Node.js, Express.js, and Flutter.
- Implemented RESTful APIs and backend functionality for retrieving and managing news content, with frontend integration through API requests.
- Built responsive and interactive interfaces with filtering and content discovery functionality, including frontend development of the Discover tab.

- Used Tailwind CSS for responsive UI development and integrated frontend and backend components into a complete application workflow.

GitHub: News Platform

Web Applications Suite **Jul 2025**

Vue.js · Node.js · Express.js

- Developed Forum, Blog, To-Do, and Reaction Timer applications with responsive interfaces and RESTful backend functionality.

Graphing & Scientific Calculator **Apr 2025**

C# · Windows Forms · OxyPlot

- Built a scientific and graphing calculator with mathematical expression evaluation, function plotting, and graphical visualization.

GitHub: Graphing Calculator

Student Management System **Mar 2025**

C++ · File Handling

- Developed a student management application supporting CRUD operations, file handling, and persistent data storage.

Technical Skills

PROGRAMMING
C, C++, C#, Java, JavaScript, Python, SQL, 8051 Assembly

WEB & SOFTWARE
HTML, CSS, Tailwind CSS, Vue.js, Nuxt.js, Node.js, Express.js, Flutter

DEVOPS & SYSTEMS
Linux/Ubuntu, Git, GitLab, GitLab CI/CD, GitLab Runner, Docker, Docker Hub, Maven, PostgreSQL, VirtualBox, Spring Boot

DATA SCIENCE & ANALYTICS
Pandas, Matplotlib, Seaborn, Scikit-learn, EDA, Statistical Analysis, Data Cleaning, Feature Engineering, Data Visualization, Machine Learning

CYBERSECURITY
Network Security, IAM/PAM, Active Directory, Vulnerability Management, OWASP, NDR, IPS, Endpoint Security, DLP, Email Security

BUSINESS INTELLIGENCE
Tableau, Excel, Pivot Tables, BI/SSBI, Data Warehousing, Data Marts, OLAP, ETL, Data Modeling, KPIs, Fact & Dimension Tables

HARDWARE & ENGINEERING
Arduino, 8051, RFID, EEPROM, SPI/Serial, LCD, Servo, VHDL, MATLAB, Proteus

DEVELOPMENT TOOLS
Postman, Visual Studio, VS Code, Android Studio, AutoCAD

Certifications

Red Hat System Administration I — Mahara-Tech / ITI — Aug 2026
Docker Training Course for the Absolute Beginner — KodeKloud — Sep 2026
Self-Presentation (US English) — Wadhwani Foundation — Jul 2026
Effective Speaking and Listening Skills (US English) — Wadhwani Foundation — Jul 2026

Education

B.Sc. Computer Engineering **Expected Jan 2028**

Arab Academy for Science, Technology & Maritime Transport (AAST) *Cairo, Egypt*

British Diploma – IGCSE **2023**

ROWAD COLLEGE *Cairo, Egypt*